

1. Introduction

The source of the dataset is the U.S. Department of Education, and it contains information about colleges in the U.S., including their location, their admission rate, the average faculty salary, and more. This data is useful because it provides aggregate data from 1996-97 through 2022-23 about many aspects of institutions in the U.S.

Our group was interested in studying whether cost of attendance had a negative linear relationship with admission rate. Since many elite private universities have high tuition rates and low acceptance rates, we hypothesized that as cost of attendance increases, admission rate would decrease. It is expected that colleges with higher admission rates would have lower cost of attendance because they are more academically and financially accessible.

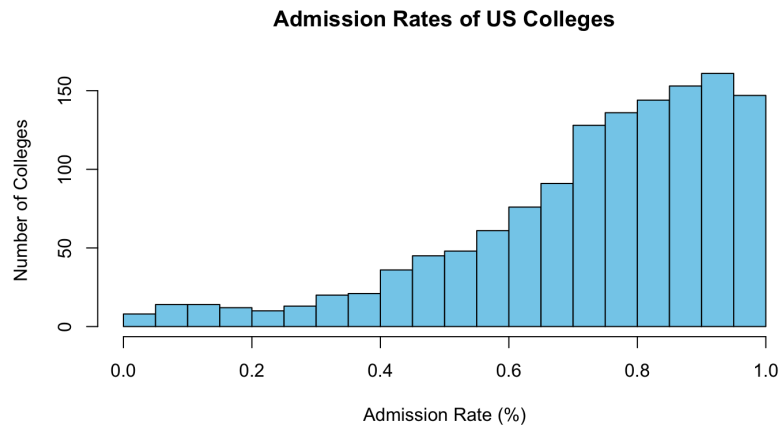
The two variables we analyzed are “ADM_RATE” (admission rate) and “COSTT4_A” (average cost of attendance). The admission rate was the independent variable and the average cost of attendance was the dependent variable. Both variables are numerical and allow us to analyze how the admission rate compares to the cost of attendance for a given university and on average. This will help us determine if there is a negative linear relationship between cost of attendance and admission rate.

Within our society, answering this research question means understanding how the cost of college might reflect the selectiveness of a university. Because many universities with lower acceptance rates are regarded as more academically rigorous, resulting in better career outcomes, our question aims to dissect how colleges with higher costs of attendance may also be more highly ranked. If colleges that are considered better academically are less financially accessible, that puts lower income students at a disadvantage.

The linear regression model is appropriate to answer our research question because it will allow us to determine the correlation between cost of attendance of admission rate. Using the linear regression model also allows us to determine the residuals of the model. We will also be able to use the correlation coefficient to understand the strength (or weakness) of the relationship between cost of attendance and admission rate.

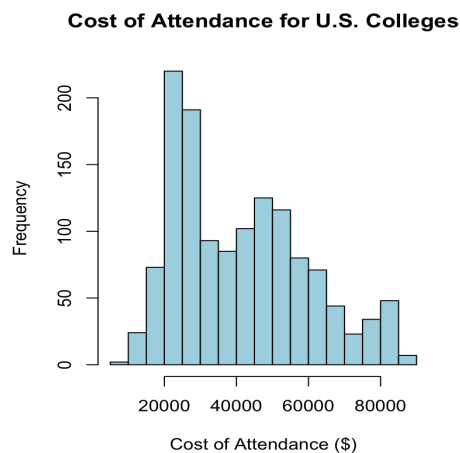
2. Data Description

The first variable our group chose for this project was the admission rate of US colleges, labelled as “ADM_RATE” in the data set. This variable is expressed as a percentage between 0 and 1. Hypothetically, a 0% admission rate would indicate that no applicants are accepted into the college, and an admission rate of 100% means that all applicants are accepted. The closer to 0 the admission rate is, the more selective the college is. The mean admissions rate of US colleges is 0.7293774, or roughly 73% percent. The standard deviation was found to be 0.2126504. Approximately 68% of the admissions rates of American colleges fall within 0.21 percentage points of the mean, which reflects a considerable variability in the admissions selectivity of the schools in the data set.

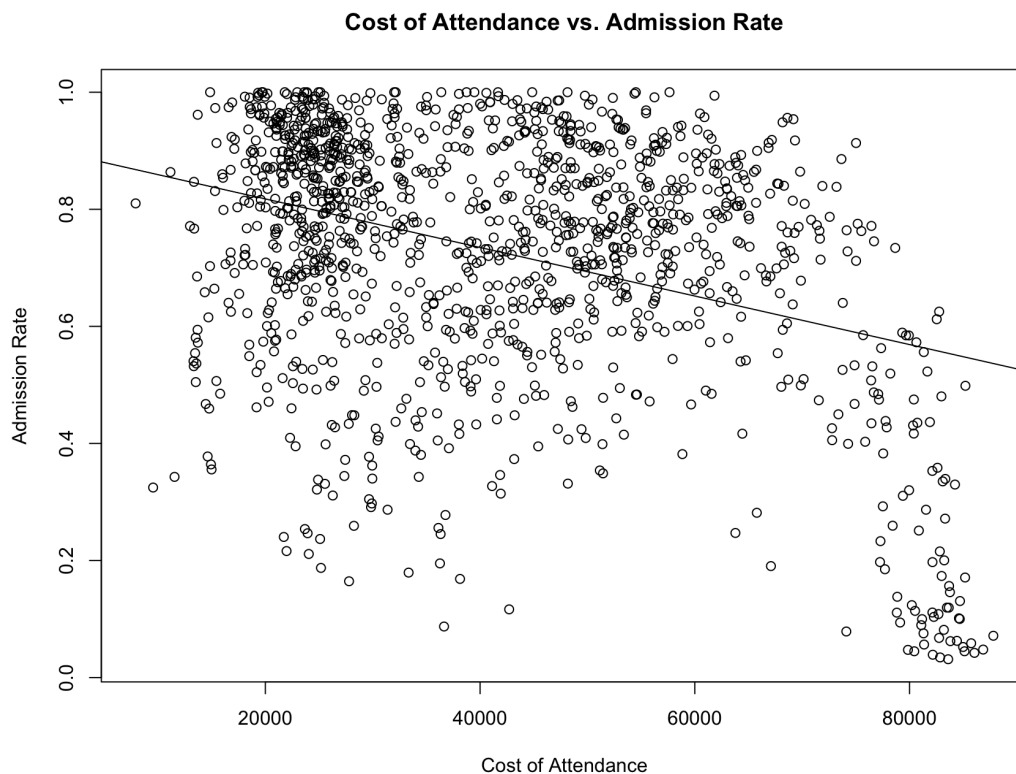


We chose to visualize the admissions rate variable as a histogram, as seen above. The data is unimodal and has a strongly left-skewed distribution, illustrating that the vast majority of US colleges have higher admissions rates while a relatively small portion are more selective. This suggests that higher education in the United States is broadly nonselective. The long tail on the left of the graph contains the colleges with acceptance rates between 0% and around 50%. The bins in this portion are very small, implying that selective admissions practices are relatively rare in the US. Overall, the histogram shows a high variability in the admissions rate of American colleges, but with a strong clustering of higher-admission institutions. This pattern could potentially reflect the landscape of higher education in the US: a small group of elite institutions contrasts with a much larger group of colleges that appeal to more applicants.

The second numerical variable our group picked is the cost for attendance (in dollars), labelled as “COSTT4_A”. The variable represents the estimated total yearly cost that a student would pay to a particular institution. The mean cost of attendance for U.S. colleges is \$41,337.37, and the standard deviation is shown to be \$18,312.36. This indicates that while the average cost of attendance for a college in the U.S. falls around forty thousand dollars a year, there is a large amount of variation from institution to institution. The high standard deviation suggests that some colleges are significantly more expensive than average (often private or proprietary), while others fall much lower, (often in-state colleges).



We chose to use a histogram to visualize the cost of attendance because this type of plot is the most effective way to show overall distribution of a numerical variable, particularly when the values span a wide range. Because college costs vary dramatically across the U.S., a histogram allows us to see how frequently different cost levels occur, as well as whether the data are concentrated in certain ranges. The histogram above reflects a right skew because a subset of institutions, typically private and proprietary, tend to have high costs compared to the average in the U.S. Although the majority of college attendance costs tend to cluster around the mid-range costs, the average is around \$41,000. Thus, it is important to keep in mind that although the mean is around \$41,000, a considerable proportion of institutions fall well below that number.



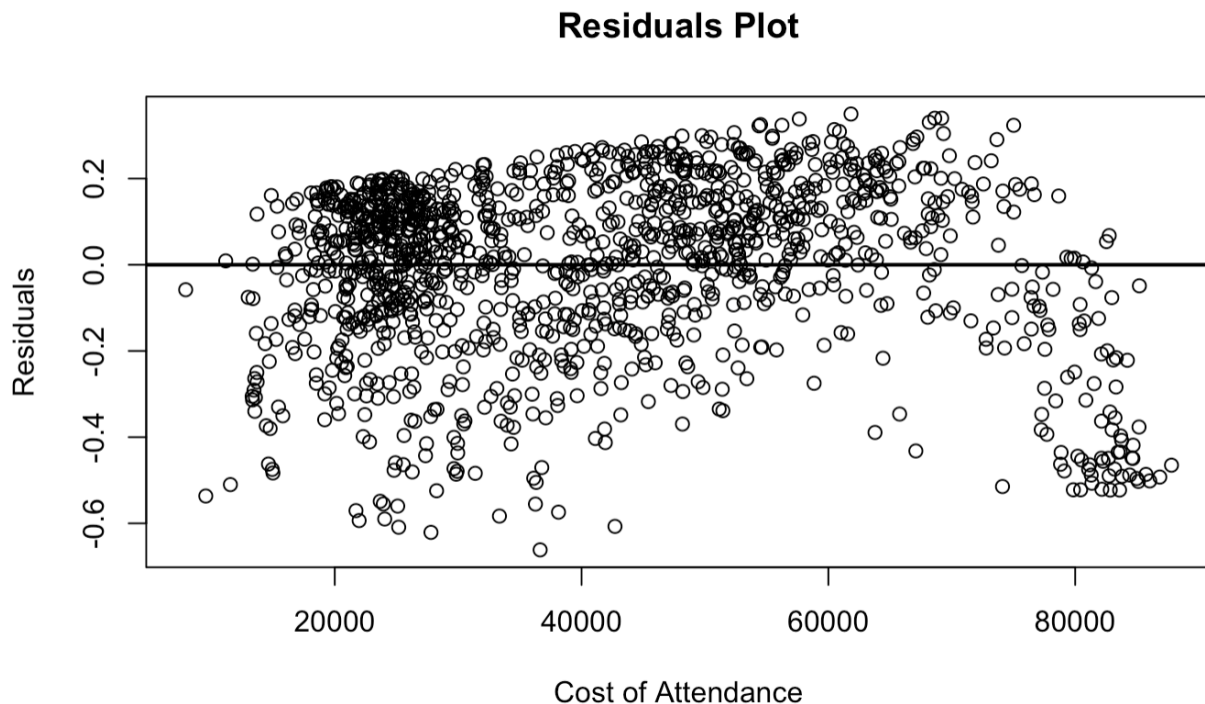
In R, we created a scatterplot comparing the admissions rate of US colleges with their average cost of attendance. The scatterplot illustrates a fairly weak relationship between the two variables, with no clear linear pattern present. There is a large, dense cluster of colleges with high admission rates and cheaper tuition, but across the full range of tuition values, colleges exhibit a wide spread of admission rates. However, there is a subtle pattern in the lower right region of the plot, where there is a group of highly selective colleges with expensive tuition rates. The correlation coefficient was calculated to be -0.357334 , indicating a weak to moderate negative relationship between the two variables. This can be visualized through the regression line overlaid on the graph.

3. Results and Interpretation

The linear regression equation for our model is:

$$y = -4.150\text{e-}06x + 9.009\text{e-}01$$

The slope is $-4.150\text{e-}06$, which means that for every \$1.00 increase in cost of attendance, admission rate decreases by 0.000415%. This is consistent with our prediction that the cost of attendance of a university would have a negative linear relationship with its admission rate. The intercept is $9.009\text{e-}01$, which means that when the cost of attendance is \$0, the acceptance rate is 90.09%. This intercept makes sense in the context of the data because the acceptance rate isn't above 100%, which would be impossible.



There is some curve to the residual plot, implying that a linear model may not be a good fit. If a linear relationship was a good model, the residuals would have no discernible pattern and instead be scattered randomly. Furthermore, the R^2 is 0.127, meaning that only 12.7% of the variance in admission rate can be explained by cost of attendance. This is a low percentage, meaning that the relationship between the two variables is not very strong.

Discussion

The purpose of this study was to determine whether the cost of attendance affects the admission rate, as most private colleges, with low admission rates, tend to have a higher cost of attendance. The

analyses presented suggested that only 12.7% of the variance in admission rate is explained by the cost of attendance. Since this is a low association, our results show a negative, weak to moderate negative relationship, indicating that as the cost of attendance increases, the admission rate tends to decrease slightly. However, since the relationship is not strong, the findings suggest that while cost may influence admission rates, many other factors also could play a much more important role in admission rate, which means our conclusions based on only cost of attendance is not significant.

These findings are reasonable in a real-world context. Many private institutions with high tuition often have lower acceptance rates due to factors such as prestige, competition, and academic rigor. However, admission decisions are influenced by many additional variables, including academic performance and enrollment goals. Because of these competing influences, cost of attendance alone does not provide a complete explanation for how selective a school may be.

However, there are limitations to our project. The data given was aggregated over a span of 26 years, which means the results do not reflect the changes that may have occurred during that time. Additionally, because the cost data are not adjusted for inflation, the values may not accurately represent the true cost of attendance in today's economy.

Overall, our results suggest that while the cost of attendance has some influence on admission rates, it is not a strong predictor on its own. This answers our research question by showing that tuition alone is not sufficient to determine how selective a college is. From our data and analysis, we can conclude that many other institutional and academic factors likely play a much larger role in explaining admission rates, and future research should include these additional variables for a more comprehensive understanding.